Stochastic Analysis

LectureNotes

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Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation then $\int_a^b f(x)\ \mathrm{d}g(x)$ is well-defined and is defined to be

$$\int_a^b f(x)\ \mathrm{d}g(x) := \lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\left(g(x_{i+1}) - g(x_i)\right)$$

with $a \le x_0 < x_1 < \cdots < x_n \le b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: Itô integral $(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$.

E.g X is Brownian motion.

Theorem 1.0.1. $\displaystyle\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1.$

Finite Variation of a function $g : [a, b] \to \mathbb{R}$ is

$$V_g([a, b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0, 1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned}(\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 &\le \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \\ &\le V_B([0, 1]) \cdot \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right|\end{aligned}$$

If $V_B([0, 1]) < \infty$, then $(\star) \to 0$.

Contradiction with $(\star) \to 1$, so w.p.1 $V_B([0, 1]) = \infty$. □

$(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$, where H_s is previsible or predictable, X_s is semi martingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$, (M_n) be a mart.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$. A stochastic process $X = (X_t)_{t\geq 0}$ is a collection of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

$$\begin{aligned} X\colon [0,\infty) \times \Omega &\to \mathbb{R}^d \\ (t,\omega) &\mapsto X_t(\omega). \end{aligned}$$

Fix ω, then the function $t \mapsto X_t(\omega)$ is called a sample path.

Definition 1.1.1. For two stochastic process X and Y, we say Y is a modification of X if

$$\forall\ t \geq 0, \quad \mathsf{P}(X_t = Y_t) = 1$$

Proposition 1.1.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

- . Assume right-continuity $\forall\ t > 0$, $\forall\ n \geq 1$, $\forall\ 0 \leq k \leq 2^n - 1$,

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall\ \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

□

Proposition 1.1.2. stopping time $\subseteq$ optional time and stopping = optional if the filtration is right-continuous.

Proof for Proposition.

- . "$\Rightarrow$" $\{T < t\} = \bigcup_{n=1}^{\infty}\{T \leq t - \frac{1}{n}\} \in \mathcal{F}_t$.
  "$\Leftarrow$" If T is an optional time, then $\forall\ m \geq 1$,

$$\{T \leq t\} = \bigcap_{n=m}^{\infty}\left\{T < t + \frac{1}{n}\right\}.$$

Because $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$.
$\Rightarrow\ \{T \leq t\} \in \mathcal{F}_{t+\frac{1}{m}}$
$\Rightarrow\ \{T \leq t\} \in \bigcap_{m=1}^{\infty}\mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t$. □

Lemma 1.1.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. □

Lemma 1.1.2. If $\{T_n\}_{n\geq 1}$ are stopping times, then $\sup_{n\geq 1} T_n$ is a stopping time.

Proof. $\forall\ t \geq 0$, $\{\sup_{n\geq 1} T_n \leq t\} = \bigcap_{n=1}^{\infty}\{T_n \leq t\} \in \mathcal{F}_t$. □

Definition 1.1.2. The stopping σ-algebra is $\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall\, t \geq 0,\ A \cap \{T \leq t\} \in \mathcal{F}_t\}$.

Proposition 1.1.3. If $T \equiv t_0$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. $\forall\, t \geq 0$,

$$\{T \leq t\} = \begin{cases} \varnothing, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

"$\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$"

$\forall\, A \in \mathcal{F}_{t_0}$, $\forall\, t \geq 0$,

① $t \geq t_0$, $A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

② $t < t_0$, $A \cap \{T \leq t\} = \varnothing \in \mathcal{F}_t$.

Hence $A \in \mathcal{F}_T \Rightarrow\ \mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.

"$\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$"

$\forall\, A \in \mathcal{F}_T$, we have $A \cap \{T \leq t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \leq t_0\} = \Omega$, so

$$A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0} \Rightarrow\ A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow\ A \in \mathcal{F}_{t_0}.$$

Hence $A \in \mathcal{F}_{t_0} \Rightarrow\ \mathcal{F}_T \subseteq \mathcal{F}_{t_0}$ $\square$

Definition 1.1.3 (defn example)**.** Example

Theorem 1.1.1 (thm example)**.** Example

Proof. proof □

Lemma 1.1.3 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.1.4. pro example

Proof for Proposition.

■. Example □

Claim 1.1.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.1.1 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.1.1. Example fact

Remark. Example remark

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